Concrete release agents

High-quality concrete release agents are indispensable where the quality of a concrete surface is concerned. Based on our decades of experience, we have developed our high-performance **RENOCAST RA** (formerly SOK) products in close cooperation with our customers. Today, these concrete release agents are utilised successfully around the world in precast plants as well as in structural and civil engineering.

RENOCAST

Application Guide: Concrete release agents in precast plants

Requirements for modern concrete release agents

Easy separation of concrete and formwork
Concrete surfaces without pores and shrink holes
Light-colored concrete surfaces without discoloration
Clean formwork
Protection of the formwork and easy cleaning
Temporary protection against corrosion on steel formwork
Hygienic safety for the user
Harmlessness for the environment
Universal applicability
Cost effectiveness
Sustainability

The fact that we can provide eco-friendly, alternative waterbased products, as concrete release agents for almost all applications clearly displays and proves that quality, efficiency and environmental awareness can be reconciled.

Our product range is also constantly evolving through our close cooperation with those people who utilise our products. Continuous dialogue with our customers is important to us. This close cooperation, coupled with our years of experience in the field of release agents, ensures that we can continue to provide ongoing product optimisation for you. In case of sufficient quantities, we will develop concrete release agents for your specific applications and requirements. Please contact us!

Sustainable construction with concrete release agents from

Release agents for highest exposed concrete requirements

The composition of concrete release agents

Concrete release agents consist of a base fluid in which the required additives are dissolved. This base fluid can be mineral oil with and without de-aromatised white spirit, ester/vegetable oil or, in the case of emulsions, water. Fatty acids and/or esters are usually used as separating substances. The fatty acids react chemically with the cations of the “concrete water” to produce metal soaps, which form the “predetermined breaking point” between concrete and formwork following the hardening of the concrete. The often utilised esters are saponified to form fatty acids and alcohol by the highly alkaline concrete (pH value 12.9). The fatty acids react, as described, with the cations in the “concrete water”. The alcohol is built into the cement matrix. Furthermore the concrete release agents contain additives for the minimisation of pores and shrink holes as well as a temporary corrosion protection for steel formwork.

RENOCAST RA 5100 C – Emulsion concentrate for civil engineering

RENOCAST MP – Concrete processing release agents

Product	Group 1 Mineral oil basis, partly with solvents	Group 2 Emulsions	Group 3 Ester-oil release agent
Carrier substance	Mineral oils / partly with solvents (approx. 80 to 95%)	Water (approx. 65 to 85%)	Vegetable oils, ester oils (approx. 95 to 99%)
Chemically separating substance	Fatty acids and esters (approx. 2 to 15%)	Fatty acids and esters	Vegetable oils, esters (approx. 95 to 99%)
Additives for minimising pores and shrink holes	yes	yes	yes
Corrosion protection additives	yes	yes	yes

Downloads

- **RENOCAST RA – Concrete release agents (Brochure)**
  1.68 MB // PDF
- **Sustainable Construction with Concrete Release Agents (Brochure DE)**
  2.75 MB // PDF
- **RENOCAST RA – Release agents for exposed concrete (Brochure DE)**
  3.03 MB // PDF
- **RENOCAST MP – Concrete mixer protection (Brochure DE)**
  1.74 MB // PDF

Other additives	-	Emulsifying agents	-

Any more questions?

Types of concrete release agents and their applications

Mineral oil based release agents

Concrete release agents based on mineral oils with and without solvents are still widely used in precast plants and in structural and civil engineering. These products are universally applicable. Precast plants often utilise solventbased products for standard formwork to fulfil the high demands on the quality of concrete surfaces (up to exposed concrete class “SB4”) and cleanliness of the formwork. The products containing mineral oil are often also used as immediate demolding release agents. According to the dangerous and hazardous substances regulations, solvent-based products must be labelled.

Your benefits at a glance:

- Application temperature up to 130°
- Reduce pore and shrink holes formation to a large extent
- Provide clean and homogeneous concrete surfaces
- Contain an effective corrosion protection
- Do not impair the adhesion of plaster, adhesives and paint
- Suitable for use in horizontal and vertical areas

Emulsion based release agents

For reasons of health and safety at work and environmental hygiene, water-based products are increasingly used in stationary production plants. Emulsions which are utilised as concrete release agents are the latest product generation. The emulsions can be used in precast plants on all standard formwork such as steel formwork, plastic-coated wooden formwork etc. Compared with solvent-based concrete release agents, emulsions have distinct advantages thanks to their handling and application. Emulsions from the RENOCAST AQUA range are non-flammable and virtually odourless during spraying. The effects of the RENOCAST AQUA products in terms of concrete technology correspond in most cases to the concrete release agents containing solvents. The user therefore obtains flawless exposed concrete surfaces that are free of pores, shrink holes and stains. The concrete surfaces are bright and smooth. Furthermore, the formwork can be easily cleaned without too much effort.

FUCHS has successfully managed to integrate a demonstrably increased corrosion protection in some products, for example, RENOCAST RA 4830 AQUA for utilisation on corrodible steel formwork.

Your benefits at a glance:

- Application temperature up to 70 °C (formwork temperature)
- Largely prevent pore formation
- Provide clean and uniform surfaces
- Contain very effective corrosion protection for steel formwork
- Do not impair the adhesion of plaster, adhesives or paint
- Suitable for use in horizontal and vertical areas
- Non-flammable
- Almost odourless in the application
- Easily biodegradability

Ester or vegetable oil based release agents

Biogenic release agents based on ester or vegetable oil are not only suitable for applications in drinking water protection areas, but can also be used in many other areas. Due to their partially high viscosity, these products can only be used to a limited extent in winter operation.

Products based on ester oil, which also include vegetable oils, are becoming more and more important for environmental reasons and are increasingly used for the production of precast concrete elements. These products are mainly found in pipe production and mixer protection, but also in the production of concrete sleepers or L-stones and the immediate demolding. FUCHS has developed RENOCAST RA 3107 , RENOCAST RA 3911 and RENOCAST RA 3210 for requirements of this kind.

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